

Pratul Muthuraja

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EDUCATION

- Illinois Insitute of Technology** Chicago, USA
Master of Computer Science January 2021 - May 2023
- Anna University** Chennai, India
Bachelor of Engineering in Computer Science June 2014 - May 2018

SKILLS

Languages	Python, Go, JavaScript, Java, SQL, Bash, HTML/CSS, Markdown
Frameworks	Django, Flask, ReactJS, NodeJS, LAMP
Tools	Kubernetes, Docker, GIT, PostgreSQL, MySQL, SQLite, MongoDB, NGINX, Helm, Gatsby, Hugo, Jenkins, ChatGPT, Elasticsearch, Grafana, Loki, Ceph, Prometheus, Ansible
Platforms	Linux, Web, Mac, AWS, GCP
Soft Skills	Leadership, Event Management, Writing, Public Speaking, Time Management

EXPERIENCE

- Eastri (Khwaash AI)** Mumbai, Remote
System Engineer October 2025 - January 2026
 - Architected a scalable AWS infrastructure supporting 100K+ users using EC2, EKS, S3, and CloudFront to ensure high availability and low-latency content delivery.
 - Designed high-level system architecture and workflow diagrams to define service interactions, data flow, and deployment topology.
 - Performed AWS cost estimation and capacity planning, forecasting infrastructure spend and optimizing resource allocation.
- Armour Chapter of Triangle** Chicago, USA
Full-Stack Developer November 2023 - November 2024

Organization Website

- Tech Stack:** React.js, Nodejs, Javascript, TailwindCSS, PostgreSQL.
- Built a full-stack web application to securely manage social and event data for over 200K members.
- Overhauled backend architecture, optimizing database queries and caching strategies to bring API response times under 100ms and reducing latency by 80% while cutting operational costs by 60%.
- Designed and implemented event-driven batch jobs for real-time notifications and automated task scheduling, improving engagement and workflow efficiency by 25%.
- Led UI/UX development, working closely with stakeholders to create Figma wireframes and translating them into a highly responsive React frontend improving load speeds by up to 40%.
- Improved system reliability to 99.99% uptime by setting up load balancing, auto-scaling, failover mechanisms, and database replication, minimizing downtime and ensuring high availability.

Private Cloud Infrastructure for Scalable EdTech Delivery

- Tech Stack:** Proxmox VE, Ceph, Kubernetes (k3s), Terraform, Ansible, ArgoCD, Helm, Prometheus, Grafana, Loki, Alertmanager, OpenTelemetry, Vault, WireGuard, Cosign, Trivy
- Set up a 3-node Proxmox VE cluster with Ceph storage to run a containerized EdTech platform supporting video lessons and interactive training modules supporting 2K+ concurrent users streaming over 8TB monthly.
- Built a CI/CD pipeline using Terraform, Ansible, ArgoCD, Helm, and k3s to streamline deployments of new course content and platform features reducing deployment time by 70%.
- Rolled out a full observability stack with Prometheus, Grafana, Loki, and OpenTelemetry to keep tabs on system health for over 200 services and handle high traffic during academic peaks.
- Locked down access for instructors and developers with Vault for secrets management and WireGuard for secure, remote connectivity decreasing incidents by 100%.
- Brought hosting costs down by 60% while maintaining 99.99% uptime, freeing up budget for content and engagement initiatives.

Internal Student Grievance & Feedback Portal

- Tech Stack:** React.js, Node.js, PostgreSQL, TailwindCSS, Framer Motion, Jest.
- Led end-to-end development of a secure feedback and grievance portal for university students, enabling real-time issue tracking and reducing resolution time by 40% through streamlined departmental workflows.
- Designed and deployed a role-based web platform using React.js, Node.js, and PostgreSQL, with full support for anonymity, notifications, and searchable case history for admins and faculty improving response rate by 35% and consequently increase engagement by 25%.

- Illinois Institute of Technology** Chicago, USA
Graduate Student Ambassador August 2021 - December 2022
 - Managed databases and generated reports on Salesforce; Analyzed prospective student data to improve response to Graduate Discover Day event resulting in 19% growth of incoming graduate students compared to 2020.
 - Strong leadership and collaboration skills evidenced by organizing recruitment activities through campus tours, student panel, and events for the Graduate Admissions Office showing 37% better response rate.